

The Challenge

You might be surprised to know that 2013 is the first year since 1987 with distinct digits. The years 2014, 2015, 2016, 2017, 2018, 2019 each have distinct digits. 2012 does not have distinct digits, since the digit 2 is repeated.

Given a year, what is the next year with distinct digits?

Input

The input consists of one integer Y ($0 \leq Y \leq 10,000$), representing the starting year.

Output

The output will be the single integer D , which is the next year after Y with distinct digits.
